Summary: Natural vs. Orpheus PT base vs. Orpheus PT steered

cs-tts-steering project

November 5, 2026

Setup. Orpheus 3B Hindi *pretrained* (`canopylabs/3b-hi-pretrain-research_release`), text-only prompt (random speaker per clip), Devanagari/English transcripts, seed 1234. Steering: CAA vector (natural minus on-policy PT, 69 train pairs) added at layer 27 attention output, $\alpha = 1$. Groups: **Natural** (TV recordings), **PT base**, **PT steered**. Section 1 is the 19 held-out test utterances; Section 2 is all 100 utterances (in-sample). Full report: `preliminary_caa_pt.pdf`.

Headline.

- Test (19): no significant gain over PT base on M1–M3; steered beats a random vector on M1 (+0.79, $p = .023$). More failures (5 vs. 1), mostly duration flags.
- All 100 (in-sample): M1 +0.84 ($p = 5 \times 10^{-5}$), M2/M3 better ($p \approx .08$), syllable rate up, voice quality worse (HNR -1.1 dB). NISQA flat.
- Objective verdict: no-go, with a weak positive signal.
- Team listening (informal): steered clips sound *a bit more natural*, but in several clips the *words are wrong or fragmented*.

Where the content errors come from. Clip-level CER (Whisper-large-v3 vs. transcript, all 100 utterances; a rough proxy):

- PT base already has poor content (test corpus CER 0.60). Steering does not make it worse overall: CER rises by more than 0.1 in 19% of utterances and falls by more than 0.1 in 28%. The exception is `train_1693_0` (steered clip 2.7× longer, CER 0.55 → 4.4).
- The transcripts often do not match the natural recordings: natural-clip CER median 0.61; 38 of 76 train utterances above 0.6. Utterances with mismatched transcripts are also where PT's output is worst (Spearman $\rho = 0.58$).
- Mismatched training pairs may also teach v "audio that does not fit the text".

Upshot. Filter the training pairs (flag mismatched transcripts, have a Hindi speaker check the worst 20 or so, rebuild v from clean pairs) or trace errors back to the transcripts and fix them at the source.

Test set, medians	Natural	PT base	PT steered
WER (median)	.688	.750	.800
Median F0 (Hz)	220	129	149
F0 range (st)	19.9	6.3	9.3
HNR (dB)	8.7	15.5	15.0
NISQA-TTS	2.94	3.62	3.57
M1 logit (mean)	5.38	−4.25	−3.85

1 Test set (19 utterances)

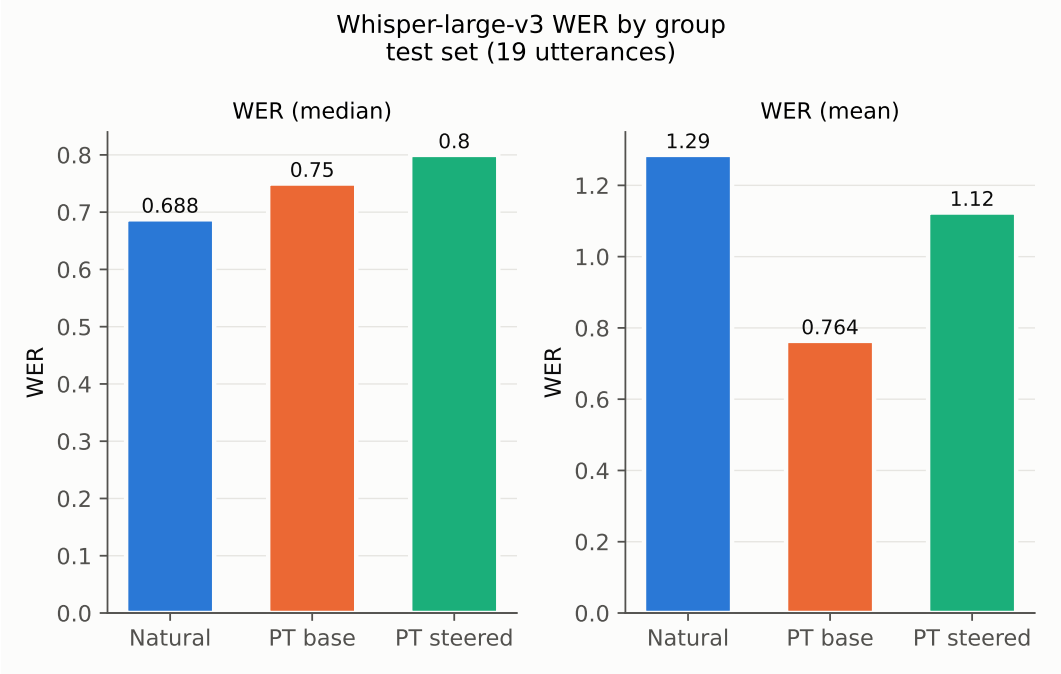


Figure 1: WER (Whisper-large-v3), median and mean. Natural WER is high, so WER is a weak intelligibility proxy here.

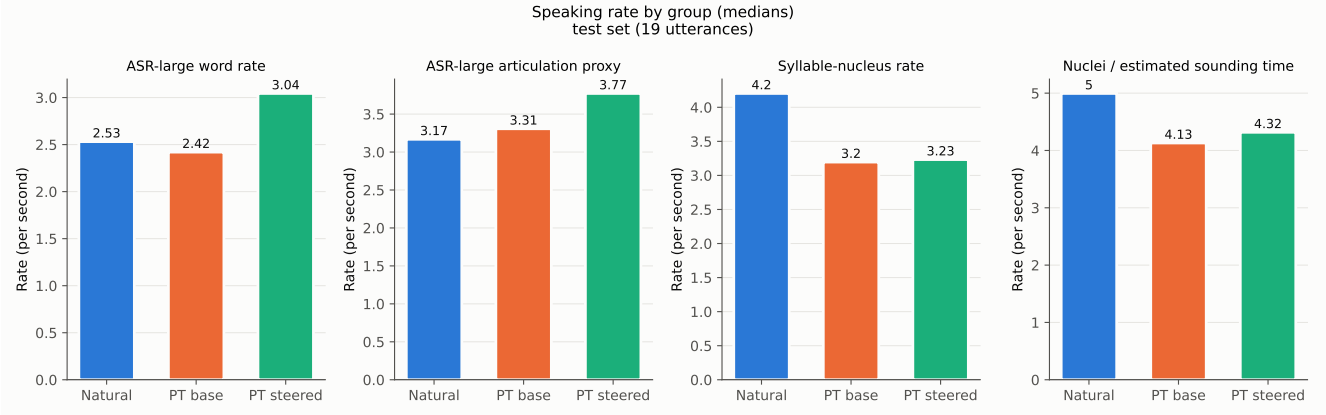


Figure 2: Speaking rate: ASR word rate, articulation proxy, syllable-nucleus rate, nuclei per sounding time.

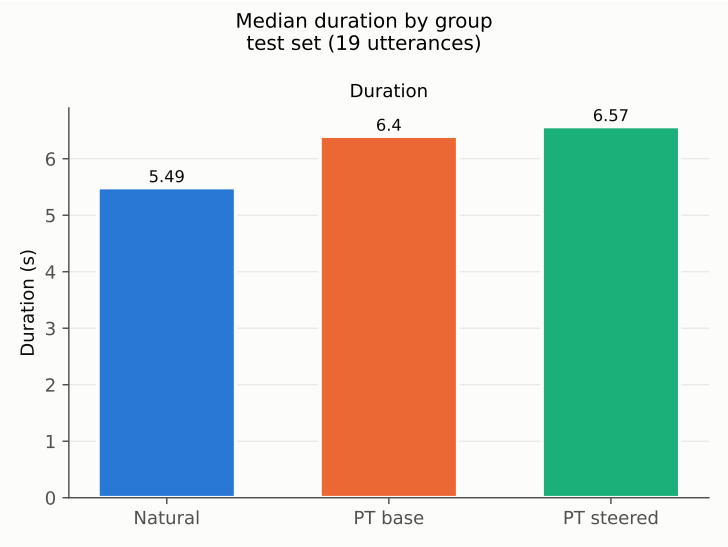


Figure 3: Median clip duration.

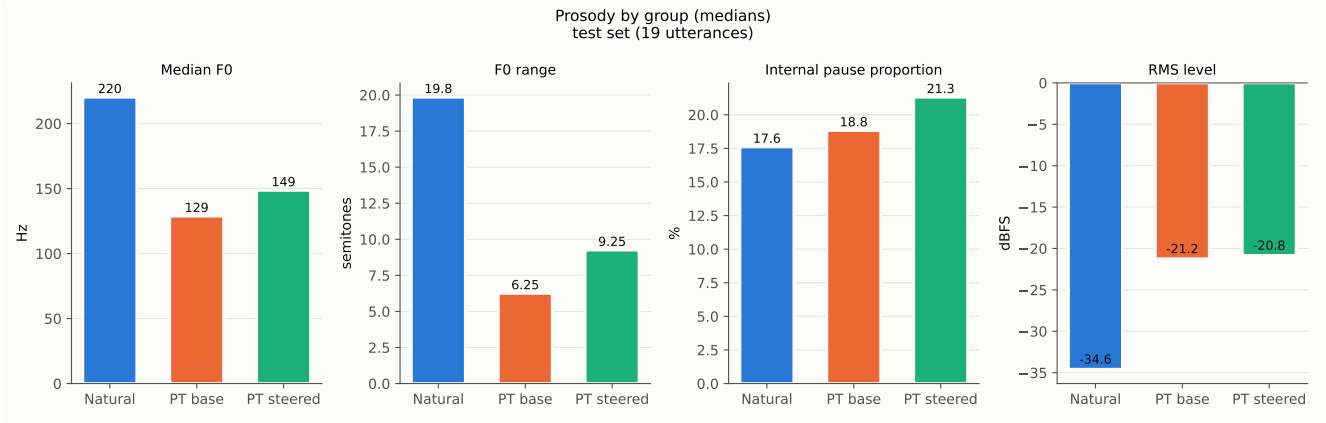


Figure 4: Prosody: median F0, F0 range, internal pause proportion, RMS level.

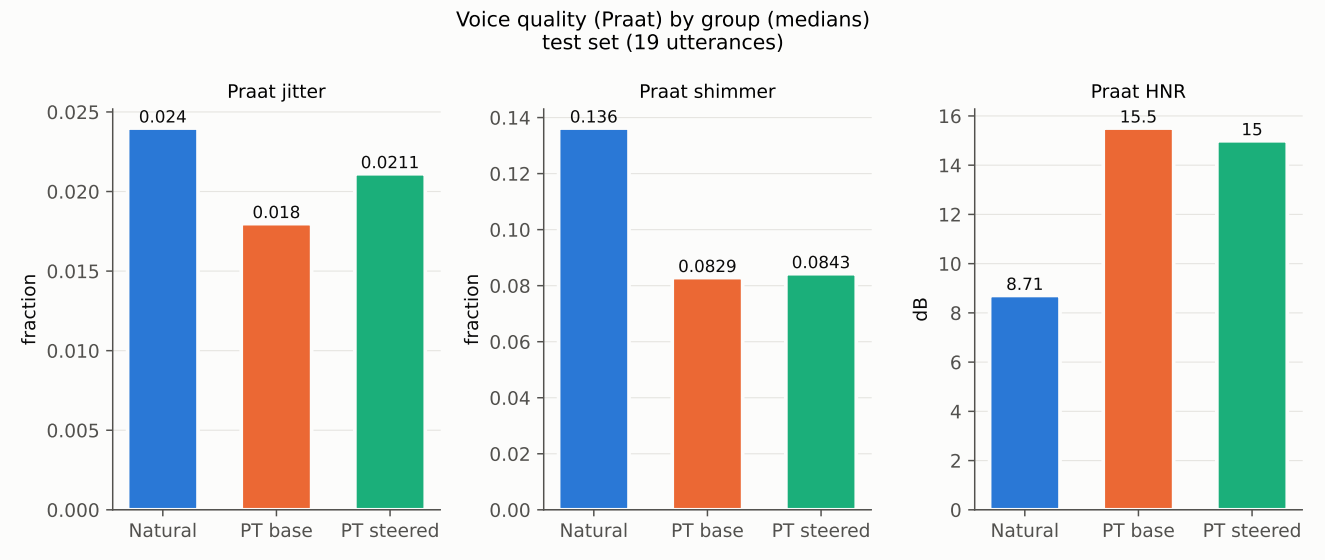


Figure 5: Voice quality: jitter, shimmer, HNR.

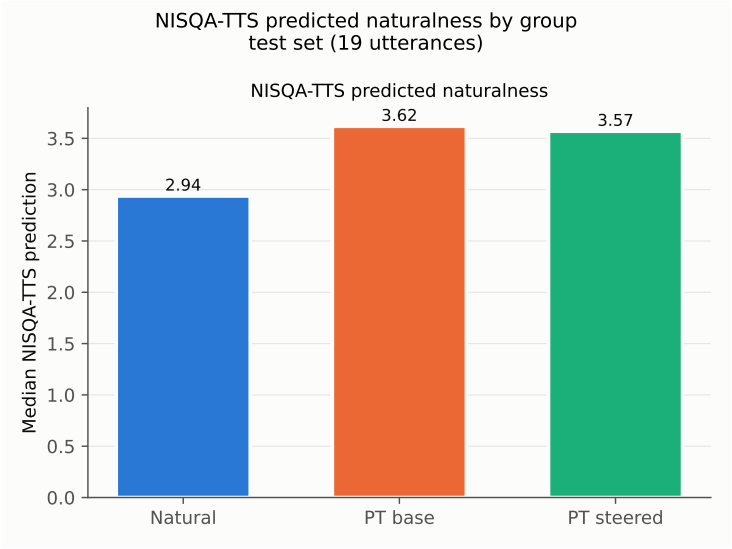


Figure 6: NISQA-TTS (learned quality score).

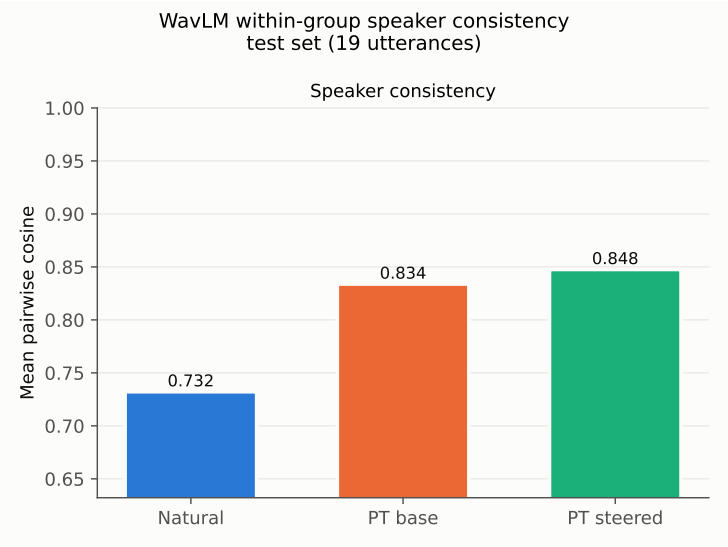


Figure 7: WavLM within-group speaker consistency (mean pairwise cosine). PT uses a random speaker per clip.

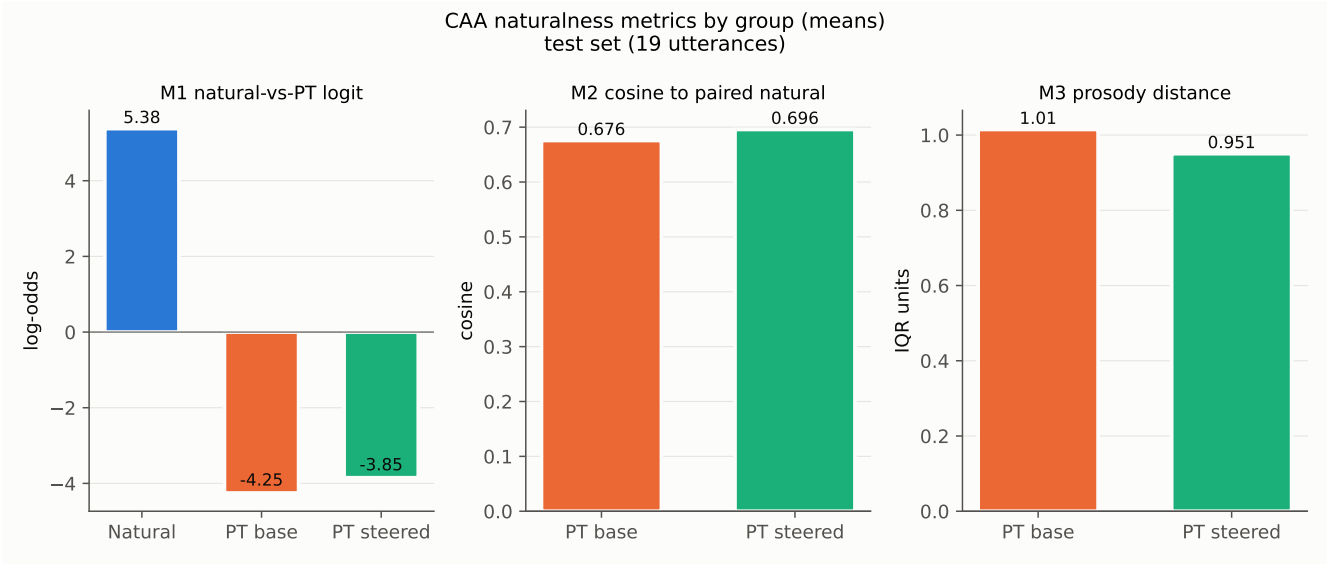


Figure 8: CAA metrics: M1 natural-vs-PT classifier logit, M2 cosine to paired natural clip, M3 prosody distance (lower is better).

2 All 100 utterances (in-sample)

The steering vector and the M1 classifier used the train utterances, so these numbers are biased in favour of steering.

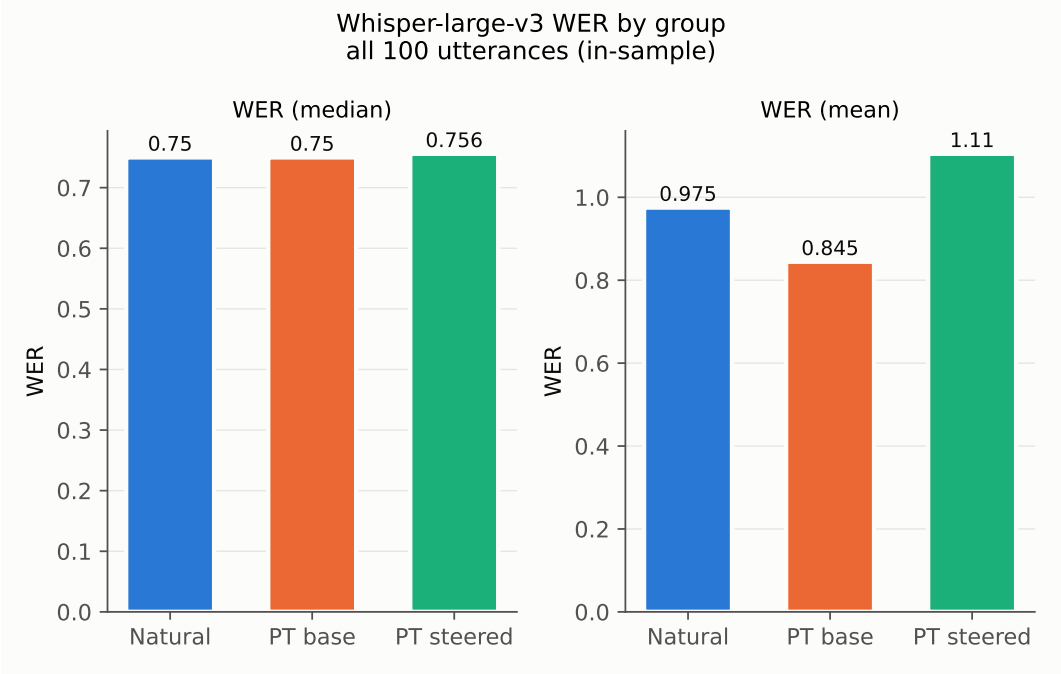


Figure 9: WER.

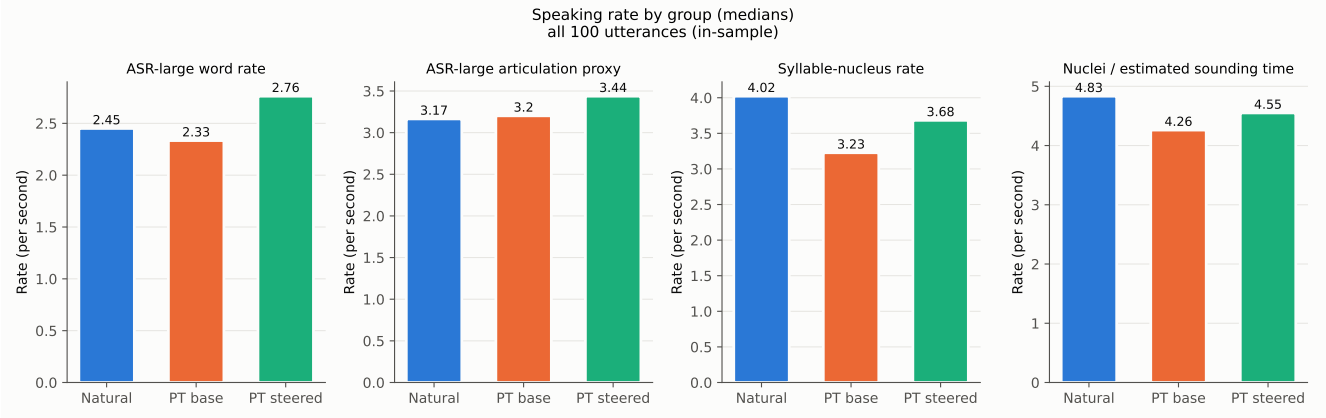


Figure 10: Speaking rate.

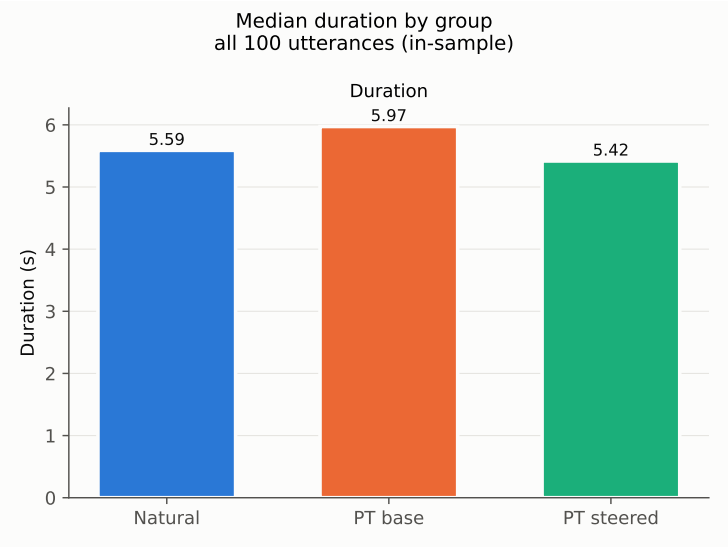


Figure 11: Median duration.

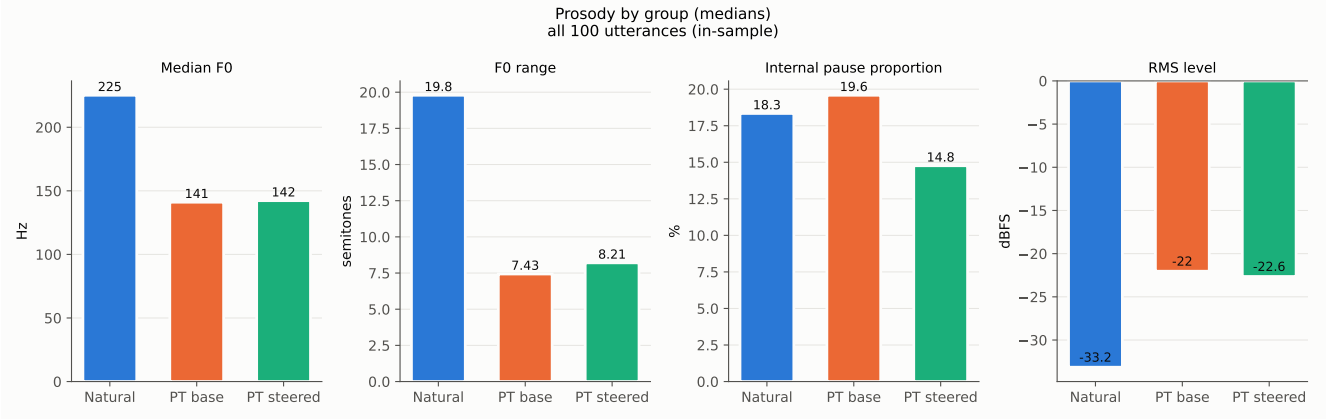


Figure 12: Prosody.

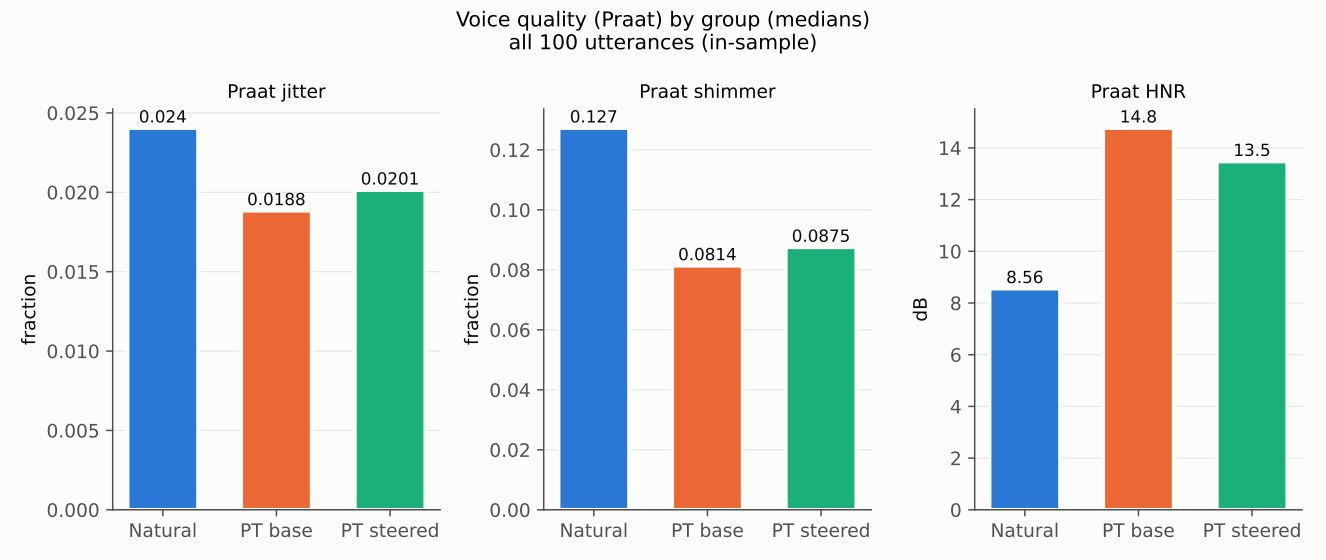


Figure 13: Voice quality.

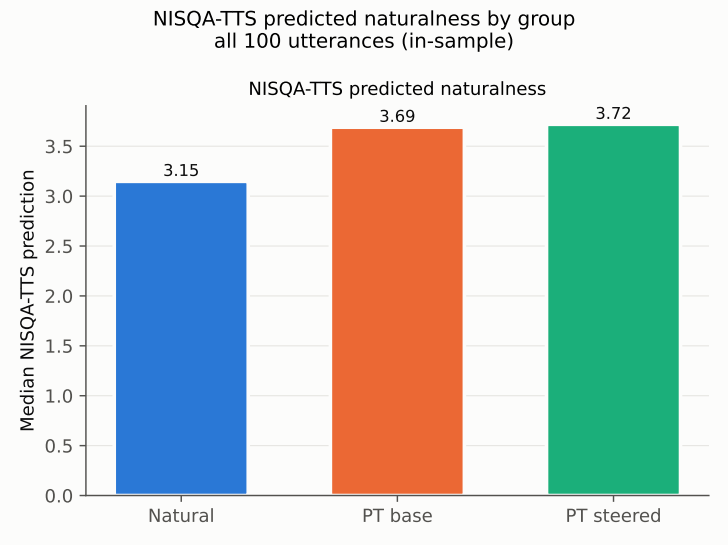


Figure 14: NISQA-TTS (one steered clip under 0.3 s unscored).

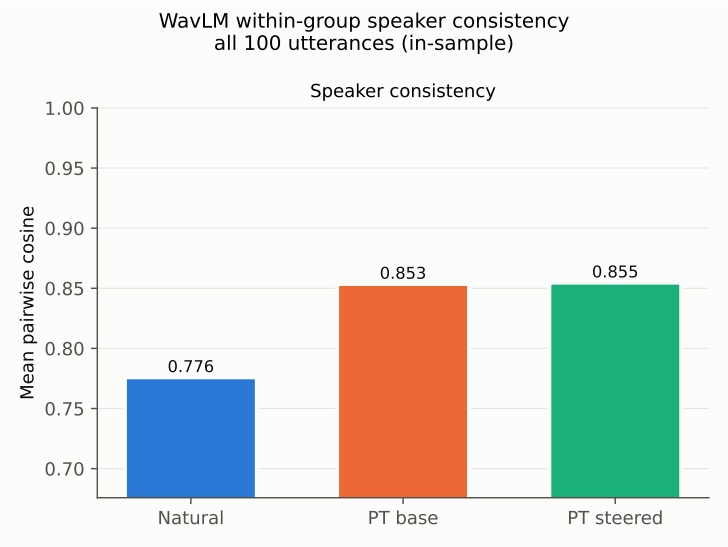


Figure 15: Speaker consistency.

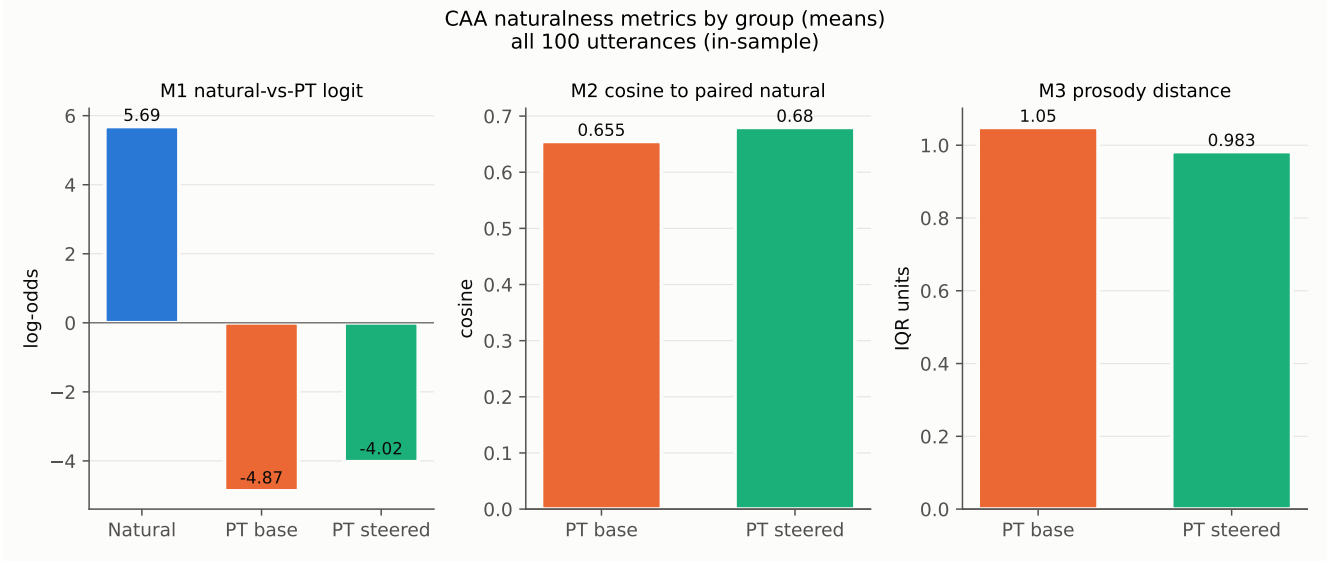


Figure 16: M1–M3.

Listening. `/local/speech/users/sp4050/Listening_Examples_100_PT/` on kedi: per utterance, the natural clip, `TRANSCRIPT.txt`, `02_OrpheusPT_base.wav`, `03_OrpheusPT_steered.wav`; open `listen.html`.